

CBSE Class 10 Mathematics — Arithmetic Progressions

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains questions. All questions are compulsory.
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. Assertion (A): The common difference of the AP $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an AP $a_1, a_2, a_3, \dots$ is given by $d = a_n - a_{n-1}$.
(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true
2. Assertion (A): The 10th term of the AP $2, 7, 12, \dots$ is 47. Reason (R): The n^{th} term of an AP is given by $a_n = a + (n - 1)d$.
(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true
3. Assertion (A): The sum of the first 10 natural numbers is 55. Reason (R): The sum of first n natural numbers is given by $\frac{n(n+1)}{2}$.
(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true
4. Assertion (A): The sequence $3, 3, 3, 3, \dots$ is an AP. Reason (R): A sequence is an AP if the difference between consecutive terms is constant.

- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
5. Assertion (A): If $2x, x + 10, 3x + 2$ are in AP, then $x = 6$. Reason (R): If a, b, c are in AP, then $2b = a + c$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

- Find the 10^{th} term of the Arithmetic Progression $5, 8, 11, 14, \dots$
- Which term of the Arithmetic Progression $21, 18, 15, \dots$ is -81 ?
- Find the sum of the first 20 terms of the Arithmetic Progression $1, 6, 11, 16, \dots$
- In an Arithmetic Progression, if the first term is 7 and the common difference is 4, find the sum of the first 10 terms.
- If the 4^{th} term of an Arithmetic Progression is 14 and the 12^{th} term is 38, find the first term a .

Section D: Long Answer (4-5 marks each)

- Find the sum of all two-digit numbers which, when divided by 4, yield 1 as a remainder.
- If the sum of the first n terms of an AP is $S_n = 3n^2 + 5n$, find the 16^{th} term of the AP.
- How many terms of the AP $24, 21, 18, \dots$ must be taken so that their sum is 78?
- Find the middle term of the AP: $6, 13, 20, \dots, 216$.
- The 4^{th} term of an AP is 11 and the sum of its 5^{th} and 7^{th} terms is 34. Find the common difference.

6. A sum of Rs. 2800 is to be used to award four prizes to students of a school for their overall academic performance. If each prize is Rs. 200 less than its preceding prize, find the value of each of the prizes.
7. The sum of the first n terms of an AP is given by $S_n = 3n^2 + 5n$. Find the 16^{th} term of this AP and the general expression for the n^{th} term.
8. In an AP, the sum of first p terms is q and the sum of first q terms is p . Show that the sum of first $(p + q)$ terms is $-(p + q)$.
9. A spiral is made up of successive semicircles, with centers alternately at A and B, starting with center at A of radii $0.5cm, 1.0cm, 1.5cm, 2.0cm, \dots$. What is the total length of such a spiral made up of thirteen consecutive semicircles? (Take $\pi = \frac{22}{7}$)